



Respiratory Procedures and Devices

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All clinical services	All clinical staff

2. Purpose

This procedure provides information and use on the various respiratory devices' patients use except for devices for manually ventilated patients.

No patient should be denied oxygen therapy in an emergency.

Oxygen should be used cautiously for patients with severe chronic lung disease and other conditions at risk of hypercapnoeic respiratory failure.

- If nebulised bronchodilators are required, the nebuliser is best driven by compressed air with oxygen therapy given concurrently via nasal cannula (without altering the current rate) to maintain an oxygen saturation of 88-92%.
- If compressed air is not available, the nebuliser can be driven by oxygen at flow rate of 6-10 L/min for three- four minutes.

Minimum infection prevention and control precautions for managing patients with an influenza-like illness must be adhered to as per the [AKG Influenza-Like Illness Guidelines](#).

Steroids especially can further narrow and even occlude the eye's drainage system so consider eye protection when nebulising Glaucoma patients.

For patient safety, the "6 Rights" of medication administration applies to prescriptions of medicated nebulisers as well.

3. Equipment

1. Choice of equipment must be appropriate for the age and/or size of the patient.
2. All equipment must be serviced and calibrated in accordance with the manufacturer's recommendations to ensure reliability and accuracy.
3. Clean and replace all equipment in accordance with the manufacturer's instructions or if visibly soiled.

4. Nebulisation

Nebulisers used by patients carrying an airborne infectious disease agent pose a risk of transmission to others in their immediate environment, so **additional precautions** are required.

Aerosol generating procedures are those that stimulate coughing and promote the generation of fine airborne particles (aerosols) resulting in the risk of airborne transmission.

To prevent errors, draw up and administer nebuliser medications separately to intravenous medications.

4.1 Patient monitoring

1. Individualised management plan to be documented in the patients' health records as soon as is practicable.
2. At a minimum the plan must consider:
 - Diagnosis.
 - Presence of comorbidities and treatment.
 - Type of monitoring e.g. 1:1, 15 minutely observations.
 - Protocol requirements.
 - Any restriction to intervention associated with advanced health directives (AHD) or the like.
3. The patient should be monitored during administration of first dose of inhaled medication or hypertonic saline nebulisation for any reaction.
 - If any wheeze or bronchospasm occurs, cease nebuliser and inform the medical officer.

4.2 Safety and precautions

- Nebulisers are a delivery system for inhaled medications or suspensions.
- The aim of nebulisation is to deliver a therapeutic dose of appropriate respirable size suitable for patient inhalation to the lower respiratory tract.
- Some nebulisers may be delivered at a lower rate (e.g. chest physio / induced sputum procedure) and can take up to 30 minutes.
- Care providers and bystanders have the risk of exposure to inhaled medications during routine monitoring and care of patients.

4.3 Clinical considerations

1. Nebuliser medications must be drawn up and administered separately to intravenous medications.
2. Prescribed nebulised bronchodilator therapy should be assessed at least daily to confirm nebulisation is indicated.
3. Nebuliser pots are for single patient use only and must be washed thoroughly or disinfected after every use as per the manufacturer's instructions.
4. If medication is to be diluted, a new ampoule of sodium chloride 0.9% (or sterile water for injection as indicated) must be used each time.
5. Each solution must be drawn up with a new needle and syringe.
6. A T-piece device is the preferred administration method.
 - Instruct the patient to maintain a tight seal with their teeth and lips around the mouthpiece.
 - Instruct the patient to breathe through the mouth as this increase's medication delivery to the lungs.
7. When using a face mask ensure a secure mask fit is maintained and adjust the metal support over the bridge of the nose.
8. Hold device in an upright position during administration.

9. If non-invasive ventilation therapy is in progress, a nebuliser can be administered if the patient can tolerate an interruption to positive pressure therapy during block care (i.e. combined nursing care).
 - If interruption cannot be tolerated, contact site specific senior nursing staff or physiotherapy staff for adapter.
 - If patient is receiving oxygen via nasal prongs, remove these for the duration of nebuliser with T piece unless otherwise instructed by medical officer.
 - If nebulisers are prescribed for every two hours, consider second nebuliser set to ensure one is always dry prior to use.
 - Check the volume fill capacity of each nebuliser set before adding prescribed medication / solution to reduce the risk of under filling or overfilling the reservoir.
 - Nebuliser chambers have a residual volume in which a small amount of liquid will remain after the treatment is finished.
 - Discard and replace the 'single patient use' nebuliser, oxygen tubing and kidney dish if visibly soiled or as per the manufacturer's instructions.
 - Provision of ongoing education and review of correct nebuliser use will help ensure the effectiveness of prescribed therapies.

4.4 Equipment

- Blunt drawing up needle (if required).
- Personal protection equipment (PPE).
- Compressed air or oxygen as prescribed.
- Prescribed solution for inhalation.
- Connecting Tubing.
- Diluent solution if nebules not available.
- Sterile syringe.
- Nebuliser set.
- T-piece nebuliser mouthpiece or mask.

4.5 Procedure

1. Assess the patient daily to confirm nebulisation is indicated.
2. Nebuliser pots are for single patient use only.
3. Use a new ampoule of sodium chloride 0.9% or sterile water for injection each time medication is diluted.
4. When nebules are not available, use a new needle and syringe for each solution to be drawn up.
5. Label the nebuliser set with the patient's identification label and date first used.
6. Explain the procedure to the patient and educate the patient, family and / or carer as able.
7. Gain verbal consent.
8. Position the patient upright, as clinically indicated to optimise lung deposition.
9. Ensure correct assembly of the nebuliser set.
10. Check the volume fill capacity of the nebuliser set before adding prescribed medication / solution to reduce the risk of underfilling or overfilling the reservoir.
11. Add the prescribed solution / squeeze contents of ampoule into reservoir of the nebuliser set.

12. Add the prescribed diluent ensuring the total volume does not exceed the recommended maximum fill volume.
 - Consider dosages to be administered as two treatments if required.
13. Discard remaining solution of diluent.
14. Secure the nebuliser cap to jar (including the baffle).
 - Ensure device is upright.
15. Secure the connecting tube to nipple of oxygen or air supply (as prescribed).
16. Turn on oxygen or air supply (as prescribed) to an adequate flow rate (may vary according to nebuliser set specifications but is usually 6-8 litres per minute)
17. Remove the nasal prongs for the duration of nebuliser unless otherwise instructed by the medical officer (MO).
18. Sit the patient upright and apply a T-piece / mask to mouth / face instructing the patient to breath steadily interspersed with occasional deep breaths.
 - If a T-piece is used, instruct the patient to maintain a good seal around the mouthpiece and breathe through the mouth.
 - Ensure a good seal with the mask and adjust the metal bridge over the patient's nose.
 - Ensure there is a mist from the T-piece or mask.
19. Nebuliser is finished when it begins to splutter (5-to-15-minute timeframe depending on volume and flow rate).
 - There may be some residual volume left in the jar.
20. Dismantle the components of the nebuliser set after use and wash the reservoir, baffle and cap with warm water.
 - Warm soapy water may be used according to manufacturer's instructions.
 - Rinse with clean water.
 - Allow to air dry on a clean surface.
21. If two hourly nebulisers are prescribed, consider a second nebuliser set to ensure one is always dry prior to use.

If non-invasive ventilation therapy is in progress, a nebuliser can be administered if the patient can tolerate an interruption to positive pressure therapy during block care (i.e. combined nursing care).

If interruption cannot be tolerated, advise the senior nurse to arrange for an adaptor.

4.6 Hypertonic saline nebulisation 3% to 14%

- Administration of a hypertonic saline (3% to 14%) may be used to increase mucociliary clearance by influencing ciliary function, increasing the osmotic drive, altering the visco-elasticity of sputum and reducing viscosity.
 - It can be used for clearing tenacious secretions in patients with cystic fibrosis and for patients post lung transplantation.

4.6.1 Safety precautions

- Ensure bronchodilator reliever therapy is readily available.
- A nebulised 3% hypertonic saline challenge is recommended for the first dose to assess possible reactivity (1.5mLs of 20% saline with 8.5mLs of sterile water).
- Patient is to be monitored during administration and observed for any adverse reactions such as chest tightness, wheeze and difficulty in breathing.

- Cease nebulisation should these occur and initiate escalation of care process specific to the site, setting and area.

4.6.2 Equipment

- Add 7.5ml sterile water for injection to 2.5ml 20% saline (total of 10mL) which constitutes 5% hypertonic saline.
- Draw up 5mL of this combined solution.
- Add to the nebuliser reservoir.
- Discard the remaining solution.
- Administer to patient.
- Refer to the previous nebulisation section for equipment and procedure.

4.7 Types of nebulisers

4.7.1 Jet

Jet nebulisers are the most common types used. They are operated by compressed air or oxygen to aerosolise liquid medications. They deliver compressed air through a jet, causing a region of negative pressure. The solution to be aerosolised is entrained into the gas stream and is sheared into the liquid film. This film is unstable and breaks into droplets due to surface tension forces. A baffle in the aerosol stream produces smaller particles.

4.7.2 Ultrasonic

Ultrasonic nebulisers convert electrical energy to high frequency vibrations using a transducer. These vibrations are transferred to the surface of the solution, creating standing waves that generate aerosol.

4.7.3 Mesh

Mesh nebulisers use electricity to vibrate a transducer (piezo) that moves liquid formulations through a fine mesh to generate aerosol.

5. Induced sputum

- Nebulisation may also be used for an induced sputum procedure which assists in the diagnoses of:
 - Pneumocystis jiroveci (PJP) previously known as Pneumocystis carinii (PCP).
 - Pulmonary Mycobacterium Tuberculosis (TB).
 - Pulmonary Mycobacterium Avium Complex (MAC).
- It may also be requested for cytology in suspected cases of lung cancer if unable to produce a sputum specimen and bronchoscopy is not considered a viable alternative.
- It is contraindicated in:
 - Pulmonary oedema.
 - Pulmonary embolism.
 - Oxygenation saturations below 90%.
 - Severe shortness of breath or bronchospasm.
 - Poor conscious level or co-operation.
 - Pneumothorax.

5.1 Equipment

- Isolated nebulizer Induced sputum kit
- 2 ampules of 6% hypertonic saline
- 10ml syringe
- 30ml 0.9% saline for gargling and rinsing mouth.
- 3 specimen jars
- Biohazard bag
- Kidney dish for mouth rinse/gargle
- Appropriate Personal protection equipment (PPE).

5.2 Procedure

1. Perform in an isolated single room.
 - A negative pressure room is required for suspected TB or MAC.
2. Performed at the request of a medical officer and requires a laboratory request form to be completed.
3. Infection control respiratory precautions must be adhered to including the use of PPE (with appropriate mask).
4. Hypertonic saline solution is administered using an Iso-Neb filtered nebuliser system
 - Inhaled hypertonic solution may cause bronchoconstriction / bronchospasm in some patients.
 - Ensure bronchodilator reliever is available for use as required.
 - Patients at risk may require bronchodilator pre-procedure.
5. Prepare patient by explaining the procedure and purpose. Gain consent from patient.
6. Patient should be in an upright sitting position unless contraindicated. Ask the patient to gargle and rinse their mouth with the 0.9% saline and blow their nose to minimize the risk of contamination of the specimen.
7. Patient instruction: "Breathe normally through your mouth and every 4-5 breaths take a deep breath and hold it for the count of 3".
8. Nebulise using 6-8L/min air via flow meter, pump or oxygen if prescribed.
9. Administer nebuliser for minimum of 10 minutes. Commence collection of the specimen after 10 mins of nebulisation. Watch for postnasal drip... blow nose if required.
10. Sputum produced may be quite watery and appear like saliva only but should still be sent to Pathology.
11. The procedure may take up to 30 minutes to complete.
12. Monitor patient during and post procedure for any adverse events and when necessary, initiate escalation of care process specific to the site, setting and area.
13. If unable to obtain a specimen, please liaise with medical officer as different concentrations of hypertonic saline greater than 6% may be required.
14. All sputum coughed to be collected in specimen containers labelled with patient's name, date, time of collection and induced sputum. Request form clearly labelled induced sputum. Specimen should be marked URGENT and send immediately to pathology.

6. Filtered Nebuliser System

- Used for the administration of medications (antibiotics or antifungal) via a nebuliser with a filter in the exhalation port.
 - This aids in the prevention of exposure of staff and others to the nebulised medications.

Pentamidine must be administered in a negative pressure room.

Staff who are pregnant or attempting to conceive should avoid exposure to pentamidine.

A filtered nebulizer system may be used for the following medications:

Pentamidine	Adrenaline	Amphotericin	Colistin
Taurolidine	Gentamicin	Mucomyst	Tobramycin
Iloprost	Pulmozyme		

6.1 Process

1. Administer in a negative pressure room.
2. A suitable well fitted one way nebuliser system should be used to reduce leakage.
3. Filter systems (either filter or tubing) must be fitted to the expiratory port to prevent environmental exposure to staff and others.
4. The nebuliser system and filter system must be used with a compatible compressor system producing an adequate flow rate and nebulisation should take no longer than 10-15 minutes to ensure maximum compliance.
5. Expiratory filters must be replaced as per manufacturers specifications.
6. A Respiratory Physician or an Infectious Disease Physician should assess whether nebulised antibiotics are indicated.
7. Consider the use of a Volumatic spacer device (holding chamber) with metered dose inhalation (MDI) administration of medication to improve efficiency by providing more reliable delivery, especially for patients who lack coordination with actuation and inhalation.
8. The medication should be administered by single actuation of the MDI followed immediately by inhalation, repeating actuation and inhalation as per the doses prescribed.
9. A test dose assessment should be undertaken in hospital to identify possible adverse reactions.
10. Perform procedure in a well-ventilated single room or isolated area whilst wearing PPE to reduce the risk of exposure to medications and droplet infection.
11. Administration through a T-piece mouthpiece is the preferred method to optimise medication delivery to the airways and reduce systemic absorption.
12. Patient education that includes correct assembly, safe use of nebuliser and potential side effects is essential to maximise therapeutic effect.

7. Nebulisation via tracheostomy tube

- Nebulisation with sodium chloride 0.9% may be prescribed as indicated for patients with tracheostomy tube insitu.
 - This should be considered in all patients using a Heat Moisture Exchange Filter (HME).
 - It is indicated when secretions are tenacious, making removal by coughing or suctioning difficult.

7.1 Equipment

- Nebuliser set.
- PPE.

- Tubing.
- Oxygen or air supply as prescribed.
- Sodium chloride 0.9% ampoule.

7.2 Procedure

1. Explain the procedure to the patient and educate the patient, family and / or carer as able.
2. Gain verbal consent.
3. Ensure patient privacy.
4. Don PPE and perform [hand hygiene](#)
5. Place 5mL sodium chloride 0.9% into bowl of nebulizer.
6. Attach the bowl to mask and tubing.
7. Connect to oxygen / air supply (as prescribed).
8. Remove garments or attachments from the tracheostomy tube.
9. Secure the mask around the patient's neck and attach to the oxygen or air source at a setting of 8-10 L/minute until nebuliser bowl is empty. Some residual volume may remain in bowl.
10. Visualise the patient throughout the nebuliser administration.
11. Document in the patient health record.

8. Inhaler devices

- These are used to deliver a variety of inhaled medications, such as beta-agonists, anticholinergics and glucocorticoids.
 - Each device has a specific formulation and dose of drug.
 - Lung deposition varies according to inhaler device, drug particle size, inhaler technique and pattern of inspiratory flow.
 - Each inhaler device has different mechanisms of operation, performance characteristics and requirements for proper use.
- Patient age, physical and cognitive ability should guide the selection of an inhaler device.
 - From infants to adults, changes in airway size, respiratory rate, inspiratory flow, breathing pattern and lung volume impact the ability to use various devices and impacts drug delivery to the lungs.
 - Physical ability, coordination (dexterity and timing) and patient preference should be considered when choosing appropriate device.
- Two main types of inhaler devices are available:
 - Pressurised metered dose inhaler (MDI).
 - Dry powder inhaler (DPI).

8.1 Pressurised Metered Dose Inhaler (MDI)

- MDIs allow low doses of medication to be delivered directly to the site of action in the airways, significantly reducing systemic side effects.
- There are two types of MDIs:
 - Conventional- press and breath design.
 - Breath-actuated-mechanism is triggered by inhalation through a breath- actuated nozzle.
- Spacers and valved holding chambers are accessory devices that reduce oropharyngeal deposition of MDI delivered drug and improve distal delivery.

8.2 Clinical considerations

Consider the use of a Volumatic spacer device (holding chamber) with MDI administration of medication to improve efficiency by providing more reliable delivery, especially for patients who lack of coordination with actuation and inhalation.

8.3 Dry Powder Inhaler (DPI)

- Administration of certain medications will require the use of a dry powder inhaler in which device specific inspiratory flow is needed to disperse the dry particles and to draw the medication from the DPI to the lung.
 - Inspiratory flow rates of 30 to 60 L/min are required to disaggregate and aerosolise medication when using a DPI.
 - They are breath-actuated devices that deliver micronized drug particles with a mass median aerodynamic diameter (MMAD) of less than 5µm.
 - Drug is delivered to the airways by the inhalation of air over a punctured capsule or blister.
- There are two main types:
- Multiple dose devices which contain up to 200 doses and the method for moving the medication containing powder into the holding chamber is either depressing a lever or twisting part of the canister e.g. Turbohaler or Accuhaler.
- Single dose devices which require placement of a capsule in the device immediately before each treatment e.g. HandiHaler, Breezhaler or Aerolizer.

8.3.1 Common errors in using inhalers

Errors when using Inhalers	Rationale for error
Failure to detect when the inhaler is empty.	Not priming as per the manufacturer's guidelines.
Poor maintenance of the device and / or spacer.	Poor coordination with actuation and inspiration.
Inspiring too fast or too slow.	Putting wrong end of inhaler in the mouth.
Failure to exhaling to functional residual capacity before inhaling or actuating at total lung capacity.	Swallowing capsules instead of putting in the device.
Exhaling into the mouthpiece of dry powdered medications.	Not shaking or tilting before use.
Tilting head excessively forward or backward.	Inhaling through the nose during actuation instead of mouth.
Cold Freon effect especially without spacer use (cold blast of propellants causes patients either to stop inhaling or inhale via the nose).	Exhaling during actuation.
Blocking the vents e.g. Turbohaler Holding mouthpiece to load instead of device e.g. Turbohaler.	Not removing the cap off mouthpiece. Floating the device in water to check if empty. Not checking the expiry dates.

9. Pressurised Metered Dose Inhaler (MDI)

- Providing verbal, written and / or physical demonstration of correct technique is essential during patient education.
- Correct technique should be regularly reinforced whenever necessary.
- Administer medication by single actuation of the MDI followed immediately by inhalation, repeating actuation and inhalation per doses prescribed.
- Consider aids that are available to enable patients with altered manual dexterity to self-administer MDIs such as MDIs with spacers e.g. Haleraid.
- Assess the patient's knowledge base and technique in using MDIs.
 - Incorrect use of inhalers reduces any potential benefit of the medication.

9.1 Instruct the patient.

1. Prime the MDI as per manufacturer's instructions.
2. Hold the MDI upright and shake well for 5-10 seconds.
3. Breathe out gently and slowly.
4. Remove dusk cap and place device mouthpiece in mouth.
5. Seal lips around the mouthpiece to create a seal.
6. Begin to breathe in slowly and deeply whilst activating the canister once only.
7. Continue slow, deep inspiration.
8. Remove the inhaler from their mouth and hold their breath for as long as comfortable.
9. If a second dose is prescribed repeat the process, waiting at least 10 seconds between puffs.
10. If required, rinse mouth post MDI administration as per medication instructions.

9.2 For use with a spacer

1. Assemble the spacer as per the manufacturer's guidelines.
2. Hold the MDI upright and shake well for 5-10 seconds.
3. Connect the MDI to the spacer and prime if necessary.
4. Breathe out gently and slowly.
5. Place lips around the mouthpiece on the spacer to create a seal.
6. Begin to breathe in slowly and deeply whilst activating the canister.
7. Take a long slow deep breath through the mouth and hold for as long as able (approx. 5-10 seconds) then breath out slowly, or
8. If the patient is short of breath, then instruct them to use the tidal technique:
 - Breath slowly and evenly in and out for four breaths (tidal breathing).
 - If a second dose is prescribed, repeat the process, waiting at least 10 seconds between puffs.

10. Dry Powder Inhaler (DPI)

10.1 Procedure

1. Instruct and demonstrate to patients in the correct use of DPIs.
2. Key points to emphasise are:
 - Storing the device at room temperature in a dry place.
 - Inhaling the medication forcefully and deeply.

- Not exhaling into the canister.
 - Not washing DPIs with soap and water as it causes the medication to clump.
 - Instructing patients **not** to swallow the capsules.
 - Instructing patients that adequate inspiratory effort is important to disaggregate the powder into fine particles and improve lung delivery.
 - Evaluating young children, the elderly and patients with acute airflow obstruction or chronic obstructive pulmonary disease (COPD) for the ability to generate a minimal inspiratory flow.
3. Consider manual dexterity devices if needed e.g. Turbogrip.
 4. Refer to the manufacturer's instructions for each DPI device technique.
 5. The patient should be instructed to:
 - Read and follow the instructions for proper assembly.
 - Make sure to keep the DPI clean and dry.
 - Keep the DPI in proper orientation during the treatment.
 - Be sure to puncture the capsule or blister pack once only.
 - **Not** exhale into the DPI.
 - Make sure to generate adequate inspiratory flow.
 - Track the devices remaining in the DPI.

11. Spirometry

- A spirometer is a device that can accurately and reproducibly measure how quickly and how much air can be moved in and out of the lungs.



11.1 Indications

- Use as an aid for preliminary diagnoses of lung disease, and also as a guide to therapy and longer-term monitoring.

11.1.1 Safety precautions

- Should only be performed by clinical staff who have had relevant education as inadequate training will result in poor quality spirometry.
- Mouthpieces must be disposed of or cleaned and disinfected between patients.
- Spirometry requires maximal effort which may cause:
 - Transient breathlessness.
 - Syncope.
 - Cough.
 - Short term oxygen desaturation.
 - Chest pain.
- May also induce bronchospasm in patients with poorly controlled asthma.

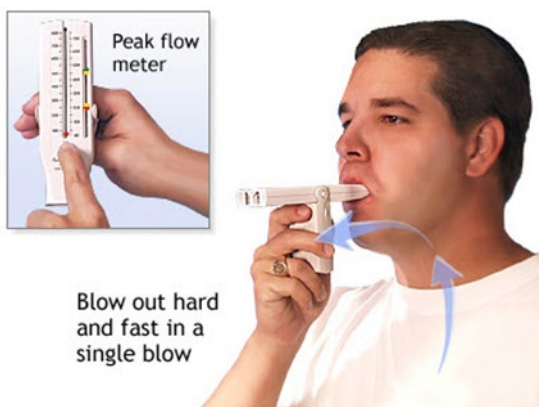
- Ensure a reliever medication is readily available during spirometry.
- Consider avoiding or delaying in the following circumstances:
 - Known or suspected respiratory infection.
 - Haemoptysis of unknown origin.
 - Pneumothorax.
 - Myocardial infarction within one week.
 - Uncontrolled hypertension or pulmonary embolism.
 - History of haemorrhagic cerebrovascular event.
 - Recent thoracic, abdominal or eye surgery.
 - Nausea, vomiting or pain.
 - Cognitive impairment.
 - Recent middle ear infection

11.2 Procedure

1. Should only be carried out by staff trained in the technique.
2. Maximum patient effort is required in performing spirometry to avoid underestimation of values.
3. Good technique is essential to ensure optimal results.
4. Calibrate devices in accordance with the manufacturer's instructions.
5. Procedure should be repeated until three acceptable and reproducible results have been obtained with a variation in FEV1 and FVC within 150mL.
6. Maximum number of attempts is generally accepted as being eight.

12. Peak flow meter

- A peak flow meter is used to measure peak expiratory flow (PEF) or peak flow.
 - This is defined as the maximum flow achieved during expiration with optimal force from maximal lung inflation.
 - Use spirometry whenever possible.



12.1 Indications

May be used:

- As a guide to the severity of airway obstruction or airway disease for a given patient at a particular time, and ongoing monitoring and initial diagnoses.
- To monitor trend, progress or deterioration of respiratory function.

- Use for trending purposes only in patients who have either been diagnosed with asthma and for whom spirometry testing is not available for whatever reason.
- To monitor response to a new treatment or change in medication and / or dose.

12.2 Safety precautions

- Inhalation or exhalation effort may stimulate the cough reflex which may affect the reading.
- Peak flow meters are single patient use only.
- Clean the device according to the manufacturer's instructions.

12.3 Clinical considerations

- Good technique is paramount for the accurate assessment of response to treatments.
 - Understanding correct technique is important to ensure accurate readings as peak flow measurements are effort and volume dependent.
- Sustained effort is not necessary as peak flow is measured in the first 10 milliseconds.
- Peak flow measurement should not be used for diagnostic purposes.
- A high peak flow does not mean the absence of lung disease.

12.4 Procedure

1. Always try to use the same meter on a given patient to maintain consistency.
2. Document when the patient last used any bronchodilator therapy.
3. Perform peak flow measurements at the same time of day with the patient in the same position as previous measurements, where possible.
4. Obtain three measurements and record the highest result as the final recorded measurement.
5. Perform peak flow both before and after inhaled medication if indicated, twice daily at least 10-15 minutes after completion of any prescribed nebulised medication.
6. Either sit the patient upright with shoulders back or patient to stand if possible.
7. Prepare the device as per the manufacturer's instructions.
8. Ensure the pointer on the peak flow meter is set at zero.
9. Patient to hold the device horizontally and avoid obstructing the pointer with their fingers.
10. Patient to take a deep breath through the mouth to full inspiration, placing their lips and teeth around the mouthpiece and ensuring a good seal.
11. Patient to blow out hard and fast in a single blow.
12. Note and document the number indicated by the pointer.
13. Reset the pointer and repeat this procedure twice more unless peak flows are not within 40L/min.
14. A 30 second interval should be given between blows.

13. Documentation

The following must be clearly documented in the patient's health record:

- **Peak flow**-record date / time and highest peak flow in health records.
- **Aerosol drug delivery**- record administration on medication chart.
 - Note any issues or difficulties in the health records.
 - Document any events specific to escalation of care in health records.
- **Spirometry** - document spirometry testing results.

- Some devices may have an electronic print out which needs to be stuck securely in the notes.

14. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Consent to Treatment Policy OD 0657/16](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching Policy EMHS: 21](#)
- EMHS [Consent Policy EMHS: 5](#)

15. Supporting information

The following documents support the implementation of this procedure:

- [Recognition and Response to Acute Deterioration AKG-POL-0388-14](#)
- [Documentation Standards – Clinical Procedure AKG-PRO-0134-11](#)
- [Influenza-like Illness Infection Prevention Guidelines AKG-GUI-0208-15](#)
- [Hand Hygiene Policy AKG-POL-0187-17](#)
- [Advanced Health Directive and Enduring Power of Guardianship Guideline AKG-GUI-0013-18](#)
- [Goals of Patient Care Policy AKG-POL-0181-18](#)
- [Personal Protective Equipment for Health Care Workers Providing Direct Patient Care Interim Guideline](#)
- National Asthma Council Australian: [Asthma & COPD Medications Chart](#)

16. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

17. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.8 Hand hygiene
- Standard 8 – Action 8.4 Recognising acute deterioration.
- Standard 8 – Action 8.6 Escalating care
- Standard 8 – Action 8.10 Responding to deterioration.
- Standard 8 – Action 8.13

18. Policy owner (relating to these procedures)

Nurse Coordinator Surgical and Acute Services

Enquiries relating to this procedure may be directed to:

Title: Nurse Coordinator

Department: Surgical and Acute Services
 Email: Shane.Hastings@health.wa.gov.au

19. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) 3 thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 2	04/06/2025	04/06/2028	Section 2 – Included mechanism of action for increased intra ocular pressure Section 2 – Included the '6 rights' of medication administration

20. Authorisation

This procedure has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Sam Jenaway – A/Deputy Director Nursing and Midwifery
Authorisation date	04/06/2025
Published date	04/06/2025